C-F-160b
March 31, 1966
SUPERSEDING
Fed. Spec. C-F-160a
September 23, 1959

FEDERAL SPECIFICATION

FEATHERS, WATERFOWL; AND DOWN, WATERFOWL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers feathers and down from domesticated waterfowl.

1.2 Classification. The waterfowl feathers and down shall be of the following types, classes, and grades, as specified (see 6.2):

Type:
- I —Feathers.
- II —Down.
- III —Feather and down mixture. (Classes and grades not applicable.)

Class:
- 1—Domesticated goose.
- 2—Domesticated duck.

Grade:
- A—Higher filling power.
- B—Medium filling power.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

V-T-276—Thread, Cotton.
QQ-S-781—Steel, Strapping, Flat.
CCC-C-429—Cloth, Osnaburg, Cotton.
CCC-C-432—Cloth, Cotton, Sheeting, (Unbleached, Bleached, and Dyed).
CCC-T-191—Textile Test Methods.
PPP-F-320—Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 148—Classification, Identification, and Testing of Feather Filling Material.
Fed. Std. No. 751—Stitches, Seams, and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-C-332—Cloth, Balloon, Cotton.
MIL-M-3985—Mothproofing Concentrate (For Textiles and Feathers).

Military Standards:

MIL-STD-105—Sampling Procedures

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and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Laboratory report approval. Unless otherwise specified (see 6.2), prior to or at the time of submission of bid, the bidder shall submit to the contracting officer a certified copy of a recent laboratory report which contains test data (see 4.2) demonstrating that the type of product proposed for delivery has been tested and found to comply with the requirements of this specification. Any of the following types of reports will be satisfactory from the standpoint of this requirement:

(*a*) An independent or commercial laboratory report.

(*b*) The prospective supplier's own laboratory report.

(*c*) A Governmental laboratory report from a contract within not more than six months of submission of bid.

The purpose of the above requirement is to assist the Government in determining the capability of the bidder to supply the specified product meeting all the applicable requirements of this specification.

3.2 Preproduction sample approval. Unless otherwise specified (see 6.2), before production is commenced, a one-pound sample of the finished product shall be submitted or made ready for the contracting officer or his authorized representative for inspection as specified in 4.3. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all the applicable requirements of this specification.

3.3 Materials. The feathers and down shall be new (not previously used or reclaimed), and shall be of such composition and grade as to yield a finished product conforming to the applicable requirements of this specification.

3.4 Processing. The materials shall be dusted, washed blood-free with a blood solubilizing agent, scoured with detergent, and rinsed. In the final rinse, a laundry sour of the fluoride type (sodium acid, silico-fluoride, or combination thereof) shall be utilized so that the finished product shall meet the acidity (pH) requirement of 3.5.3. The feathers shall be subsequently dried and sterilized (see 6.3).

3.5 Finished requirements.

3.5.1 *Oxygen number.* The finished product shall show an oxygen number not greater than 10 when tested as specified in 4.7.

3.5.2 *Solvent soluble material.* The solvent-soluble material in the finished product shall not exceed 2.5 percent when tested as specified in 4.7. When DDT is required (see 3.5.8), the solvent-soluble material exclusive of DDT shall not exceed 2.5 percent.

3.5.3 *Acidity* (pH). The finished product (or product prior to application of DDT) shall show a pH value within the limits of 3.5 to 6.0 when tested as specified in 4.7.

3.5.4 *Brittleness.* The whole feathers components in all types shall be pliable and shall show no more than 8.0 percent failures (4 fractured feathers), when tested as specified in 4.7.

3.5.5 *Composition.* The finished product shall conform to the applicable requirements for composition in table I when tested as specified in 4.7. Classification of components shall be in accordance with the applicable definitions in Fed. Std. No. 148.

3.5.6 *Filling power.* The finished product shall conform to the applicable requirements for filling power in table I when tested after two launderings as specified in 4.7.

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TABLE I. *Composition and filling power*

	Type I		Type II Classes 1 and 2	Type III
	Class 1	Class 2		
Major classification (percentage of total, by weight):				
Pure down (and plumules)	10 min.	10 min.	80 min.	40 min.
Pure down (plumules) and whole waterfowl feathers (cumulative)	90 min.	90 min.	—	88 min.
Extraneous (total of components below)	10 max.	10 max.	20 max.	12 max.
Extraneous component (percentage of total, by weight):				
Small waterfowl feathers	—	—	15 max.	—
Nonwaterfowl feathers	8.0 max.	8.0 max.	[1]	4.8 max.
Feather fibers	[1]	[1]	5.0 max.	[1]
Down fibers	1.0 max.	1.0 max.	8.5 max.	4.0 max.
Quill and damaged feathers (see[1] references)	5.0 max.	5.0 max.	1.0 max.	5.4 max.
Residual matter	2.0 max.	2.0 max.	2.0 max.	2.0 max.
Feather lengths (percentage of whole waterfowl feathers, by weight):				
Less than 2.5 inches	—	50 min.	15 max.	—
Less than 3.0	50 min.	—	—	50 min.
Less than 3.5 (cumulative)	—	95 min.	—	—
Less than 4.0 (cumulative)	95 min.	—	—	95 min.
3.5 to 5.0	—	5 max.	0.0	—
4.0 to 6.0	5 max.	—	0.0	5 max.
5.0 and over	—	0.0	0.0	—
6.0 and over	0.0	0.0	0.0	0.0
Filling power (cm) after launderings (2)				5.7
Grade A	5.7	4.8	8.5	
Grade B	4.8	3.8	7.0	

[1] To be included with quill and damaged feathers.

3.5.7 *Odor.* The finished product shall have no odor of putrefaction when tested as specified in 4.7.

3.5.8 *Insecticide.* When specified (see 6.2), the feathers and down shall be treated with DDT (dichlorodiphenyl-trichloroethane). The DDT shall be applied from an aqueous emulsion in accordance with "Direction for Use (Feathers)" of MIL-M-3985. The finished product shall contain not less than 0.20 nor more than 0.70 percent of DDT based on the dry weight of product when tested as specified in 4.7.

3.6 **Weight basis.** When specified (see 6.2), or in case of dispute, weight of product shall be based on standard weight at equilibrium moisture content under standard atmospheric conditions as defined in Fed. Std. No. 148. Standard weight shall be determined as specified in 4.7.3.1.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Laboratory report approval. The data contained in the report submitted in accordance with 3.1 shall be obtained in accordance with test methods specified in 4.7.

4.3 Preproduction sample inspection. The preproduction sample submitted in accordance with 3.2 shall be examined and tested to determine compliance with the requirements of section 3.

4.4 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5 In-process inspection. Inspection shall be conducted to see that the processing is in accordance with the specified requirements. The supplier shall certify conformance to the following requirements:

a. Use of new feathers and down, as applicable.
b. Conduct of processing as specified (see 3.4).

4.6 Examination of bags used for packaging. The sample unit for this examination shall be an unfilled bag turned inside out. The bags shall be examined to determine conformance to the construction and workmanship of the inner seams as specified in 5.1. The inspection level shall be S-2 and the acceptable quality level (AQL) shall be 4.0 defects per hundred units of MIL-STD-105.

4.7 Testing of the end item. Testing of the end item shall be performed in accordance with the methods specified in table II. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The test methods indicated by number shall be performed as specified in Fed. Std. No. 148. The sample unit for testing shall be a thoroughly mixed 24-ounce composite of the finished product. Not more than 6 ounces shall be taken from any level A (see 5.1.1) package (bag). For larger bags, samplings shall be from top, center, bottom, and opposite sides of the bag, with no more than 12 ounces from any one bag. Care shall be taken in mixing to prevent damage or compression of the material. The lot size shall be expressed in terms of pounds each. The sample size (number of sample units) and the acceptance number shall be in accordance with the following:

Lot size (pounds)	Sample size (units)	Acceptance number
3200 or less	2	0
3201 up to and including 10,000	3	0
10,001 up to and including 35,000	5	0
35,001 and over	8	1

TABLE II. *Test methods* [1]

Characteristic	Requirement paragraph	Test method
Oxygen number	3.5.1	4
Solvent soluble material	3.5.2	5
Acidity (pH)	3.5.3	8
Brittleness	3.5.4	3
Composition	3.5.5	2 and 4.7.1
Filling power after laundering	3.5.6	1 and 4.7.2
Odor	3.5.7	11
DDT content (when specified)	3.5.8	6

[1] Testing may be performed under prevailing atmospheric conditions except in settlement of dispute in which case standard conditions as defined in Fed. Std. No. 148 shall apply.

4.7.1 *Components.* Components shall be classified and weighed in accordance with the applicable categories in table I and as defined in Fed. Std. No. 148. Whole waterfowl feathers shall be separated into the applicable length ranges indicated in table I, and their percentages by weight reported individually or cumulatively, as applicable.

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4.7.2 *Laundering procedure.* Place approximately 1 ounce of the composite sample in a 17- by 6-inch cloth bag made of down-tight but water-permeable cloth such as type I of MIL-C-332. The bag shall be tightly seamed and closed to prevent loss of material during laundering. The filled bag shall then be subjected to two successive complete launderings (wool method) in accordance with the general procedure of method 5556 of CCC-T-191. The dryer shall be adjusted to have a stack temperature of 150° to 170°F. After the two wash cycles, the material shall be removed from the bag, conditioned and tested for filling power in accordance with method 1.

4.7.3 *Examination of the end item.* The end item (completely packaged and packed product) shall be inspected in accordance with the examination and tests specified in 4.7.3.1, 4.7.3.2, and 4.7.3.3.

4.7.3.1 *Weight.* The lot shall be unacceptable if the total net weight of the contents in the sample is less than the total net weight indicated on the bags. The number of bags to be taken for the weight examination shall be in accordance with level S-4 of MIL-STD-105.

4.7.3.1.1 *Standard weight basis.* When specified, or in case of dispute (see 3.6), the weight basis for acceptance shall be determined by correction of unconditioned weight, to weight at equilibrium under standard atmospheric conditions as defined in Fed. Std. No. 148. At the time of weighing, a random sample of product not less than 1/4 pound shall be taken from the contents of the bag, and placed immediately in a tared closable container. The sample shall then be weighed in the closed container. The sample shall then be spread in a shallow layer in a tray and stored in the standard atmosphere for not less than 24 hours, with occasional stirring. The conditioned sample shall then be weighed, and the gain or loss calculated to a percentage based on the original sample weight. The percentage correction shall then be applied to the unconditioned weight of the contents of the bag to determine its weight on the standard weight basis.

4.7.3.2 *Packaging examination.* The sample unit shall be one bag of product. The inspection level shall be S-1 of MIL-STD-105 and the AQL shall be 6.5 defects per hundred units. Defects shall be scored in accordance with the list below.

Defects
Not packaged in accordance with specified packaging level.
Bag not new, or any patch.
Not closed as specified.
Wire tie broken or not tightly turned to accomplish a secure closure.
Any hole, rip, tear, open seam; any weak places in bag which may develop into a hole; backtacking and backstitching not as specified.
Bag wet.

4.7.3.3 *Packing examination.* The sample unit shall be one bale. The inspection level shall be level I of MIL-STD-105 and the AQL shall be 6.5 defects per hundred units. Defects shall be scored in accordance with the list below.

Defects
Not packed in accordance with specified packing level.
Bags not compressed to specified size.
Does not contain the specified number of bags.
Kraft facing of baling board not on outside.
Any board missing or strap missing.
Baling board or strap too loose or not placed in proper location to a degree where safe delivery of contents may be impaired.
Marking—missing, incomplete, illegible, incorrect, not as specified, or not in proper location.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.* Fifty ± 5 pounds of type I or III, or 25 ± 3 pounds of type II product, shall be packaged in a new (not used or laundered) plain textile bag fabricated from cotton sheeting conforming to type

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VIII of CCC-C-432, or osnaburg conforming to class 2 of CCC-C-429. Fabrics in other widths of equivalent weight and construction will be acceptable.

5.1.1.1 *Bag, construction.* The bags shall have a finished length and width, measured flat, of approximately 72 by 36 inches and shall be constructed in accordance with the following requirements: The side and bottom shall be seamed with 5 to 6 stitches per inch in seam type SSa-2, with either stitching type 401 or 301, conforming to Fed. Std. No. 751. The stitching shall be at least 5/8 inch from the cut edge of the fabric or 1/4 inch from a selvage edge, and the rows shall be at least 1/8 inch apart. The ends of all stitching, when not caught in other seams or stitching, shall be backstitched or backtacked at least 1 inch. The thread used for seaming shall be cotton, machine type IA1 or IB1, unbleached, ticket No. 12, 4 ply conforming to V-T-276.

5.1.1.1.1 *Type* 301 *stitching.* Thread breaks in stitching shall be backstitched not less than 1 inch at each break. Thread tension shall be maintained so that there will be no loose stitching. Thread ends shall be trimmed.

5.1.1.1.2 *Type* 401 *stitching.* Thread breaks in stitching shall be overstitched. not less than 1-1/2 inches at each break. Thread tension shall be maintained so that there will be no loose stitching. Thread ends shall be trimmed.

5.1.1.2 *Closure.* The closure of the top of the bag after filling shall be effected by means of two double loops of 0.072 inch thick galvanized soft iron or steel wire ties of not less than 6 inches long, having a 1/2 inch diameter eye formed at each end. One tie shall be applied at the base of an ear formed by gathering the cloth evenly and the second tie applied at a distance approximately 1 inch from the first tie with the twisted ends of two ties alternated and facing in opposite direction.

5.1.2 Feathers and down shall be packaged to afford adequate protection against damage and deterioration during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 *Level A.* Three bags of type I or III product, or four bags of type II product (see 5.1.1) shall be placed between two baling boards and compressed by machine so as to form a bale approximately one-fifth of the original volume. Bales thus formed shall measure approximately 60 by 24 by 20 inches.

5.2.1.1 *Baling boards.* Baling boards shall be not less than 0.075 inch thick solid fiberboard conforming to type SF, class domestic, grade 200 of PPP-F-320, except that at least one face of the board shall be not less than 0.010 inch kraft paperboard. Two such baling boards, size 58 by 36 inches, each with a score line placed 6 inches from each 58-inch edge in such manner that the kraft facing is on the outside when in a folded position, shall be used in each bale; one board on top of bags being baled, and one on the bottom. The boards shall be bent at the score lines thereby forming a fiberboard cap 58 inches in length, 24 inches in width, with 6-inch flanges extended down over the side of the bale.

5.2.1.2 *Strapping.* Not less than eight straps shall be used on each bale; straps shall be affixed girthwise around the sides, top, and bottom of the bale over the baling boards. The two outside straps shall be placed 4 inches in from each end and the intermediate straps placed equidistant between the end bands. All strapping shall be flat steel not less than 3/8 inch wide by 0.020 inch thick conforming to type I, class A of QQ-S-781.

5.2.2 *Level C.* Feathers and down, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest

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transportation rates for such supplies. Containers shall be in accordance with the rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Type I covered by this specification is intended primarily as filling material for pillows. Types II and III are intended primarily for sleeping bags.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, and grade (as applicable) of product required (see 1.2).
(*c*) When DDT treatment is required (see 3.5.8).
(*d*) When laboratory report approval is not required (see 3.1).
(*e*) When preproduction sample is not required (see 3.2).
(*f*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(*g*) When weight for acceptance is on standard weight basis (see 3.6).

6.3 Processing. With reference to the processing requirements of 3.4, the following procedure is suggested:

a. Feathers or down are agitated for 20 minutes in water at 95° ± 5°F. to which has been added 1 ounce of blood-solubilizing agent for each 10 gallons of water.

b. Material is then scoured with detergent in warm water and rinsed until the rinse water is substantially clear. In the final rinse, a commercial sour of prescribed type is added.

c. *Dry heat sterilization.* Sterilization by dry heat requires developing a holding temperature of not less than 165° F. for 5 minutes in an approved drying chamber with constant agitation. Each drying chamber shall be equipped with a temperature measuring device so located within the dryer exhaust area to insure that all feathers have been subjected to at least 165°F. for not less than 5 minutes. The temperature measuring device must be attached to a continuous recorder for record purposes. Since temperature gradients will exist within the drying chamber, care should be taken to locate the temperature measuring device in the exhaust area so that the discharged dryer air temperature will give a fair indication of the degree of heat the feathers received during the drying cycle.

d. *Chemical sterilization.* If these conditions cannot be met, feathers should be soaked in a solution of 2000 parts per million Disinfectant, Germicidal and Fungicidal, Concentrate (Phenolic, Dry-Type) MIL-D-51061, FSN 6840-753-4797-by the following process:

The washed feathers are immersed in water at 90° to 100°F. There should be enough water to cover the feathers. Sufficient disinfectant is added to bring the concentration to 2000 p.p.m., (about 26 ounces per 100 gallons). The feathers are held in this solution for 15 minutes with occasional agitation after which the solution is drained. The feathers are centrifuged and dried in the usual manner.

In this process, the following precautions must be maintained with use of referenced disinfectant:

(1) Contact of dust or solution with the eyes and skin should be avoided by the judicious use of gloves, aprons, face shields, and/or safety goggles.

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(2) Where exposure of personnel to the disinfectant in dust form is likely, toxic-dust respirators should be used.

MILITARY CUSTODIANS:

Army—GL
Navy—SA
Air Force—67

Review activities:

Army—MD
Navy—SA
Air Force—67

User activities:

Navy—MC, MS
Air Force—26
DSA—DM

Preparing activity:

Army—GL

Review and user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION: Feathers, Waterfowl: and Down, Waterfowl – C-F-160b

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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